EV Charging Network

Performance & Business Analytics

Analytical Report: Findings, Insights and Recommendations

100,000 charging sessions | 8 Indian cities | Jan 2024 - Dec 2025

Tools: Python (pandas, Matplotlib, Seaborn) and Power BI

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Contents

1	Executive summary	3
2	Background, objectives and scope	4
3	The dataset	4
4	Method and tools	5
5	Data quality and cleaning	6
6	What the cleaning left behind	7
7	Headline KPIs	9
8	Power BI dashboard walkthrough	10
9	When people charge	12
10	Seasonality and year-on-year	13
11	City analysis	14
12	Charger types	15
13	Vehicle types	16
14	Station types and stations	17
15	Payment behaviour	18
16	Revenue drivers and pricing	19
17	Wait time	20
18	Customer ratings	21
19	Consolidated insights	22
20	Recommendations	23
21	Limitations, next steps and conclusion	25

1. Executive summary

What the data says in one page

100,000 Charging sessions	**Rs 82.3M** Total revenue	**5.15M kWh** Energy delivered	**Rs 823** Revenue per session
23.3 min Average wait	**64.9 min** Avg charge duration	**51.5 kWh** Avg energy / session	**3.00 / 5** Average rating

This report analyses 100,000 EV charging sessions recorded across eight Indian cities, 300 station IDs and four charger types between January 2024 and December 2025. It combines the Python cleaning and exploration notebook with the two-page Power BI dashboard, and goes further by testing what the numbers can and cannot support.

The five things that matter most

- **The network is extremely evenly balanced.** Cities, vehicles, stations, payment modes and weekdays all sit within about 1-2% of each other, so no segment is a risk and none is a clear star.
- **Two apparent leaders are probably cleaning artifacts.** Pune (about 2,080 extra sessions) and the AC 7kW charger (about 2,300 extra) lead because missing values were filled with the most common category. They should not be treated as real strengths until the raw data is checked.
- **Revenue is driven by energy, nothing else.** Revenue tracks kWh delivered (r = 0.89) at about Rs 16 per kWh. City, charger, vehicle and payment mode do not move it.
- **Wait time is a network-wide problem.** The average wait is 23 minutes and 90% of sessions wait up to 41 minutes. It is nearly identical in every city, hour and station type.
- **Ratings do not respond to anything.** They are spread evenly from 1 to 5 stars and are unrelated to wait time, so they cannot yet guide decisions.

> **Bottom line:** this is a healthy, steady network earning Rs 82.3M. The biggest opportunities are shorter waits, trustworthy customer feedback, and cleaner data, not shifting investment between cities or charger types.

2. Background, objectives and scope

EV charging is a capital-heavy business. Operators need to know where sessions happen, how much each one earns, how long customers wait and whether they leave satisfied. This project turns a session-level dataset into that kind of picture, for a network that spans Ahmedabad, Bengaluru, Chennai, Delhi, Hyderabad, Kochi, Mumbai and Pune.

Objectives

- Clean and validate the session data so that every number in the dashboard can be trusted.
- Measure the headline KPIs: sessions, revenue, energy, wait time, charging duration and rating.
- Compare performance across city, charger type, vehicle type, station type and payment mode.
- Find the drivers of revenue and of customer experience.
- Turn the findings into specific, prioritised recommendations.

Scope

The analysis covers the 100,000 cleaned sessions in EV_Cleaned.csv. It has no station capacity, tariffs, costs or customer identities, so profit and utilisation cannot be calculated.

3. The dataset

One row is one charging session

Column	Meaning	Range / values
Session_ID / Session_Start	Unique number; start date and time	1 to 100,000; Jan 2024 to Dec 2025
City	City of the session	8 cities
Station_ID	Station code	300 codes (STN-xxx)
Station_Type	Location category	Highway, Mall, Office, Residential, Urban
Charger_Type	Charger power class	AC 7kW, DC 30kW, DC 60kW, DC 120kW
Vehicle_Type	Vehicle charged	Bike, Bus, Car, SUV, Taxi
Energy_kWh	Energy delivered	8.0 to 95.0 kWh
Wait_Time_Min	Time in queue	0 to 219 min
Charging_Duration_Min	Time on charger	10 to 120 min
Revenue_INR	Amount billed	Rs 80.79 to Rs 2,082.29
Customer_Rating	Star rating	1 to 5
Payment_Mode	How the customer paid	Card, UPI, Wallet, Fleet Account
Hour, Day, Month	Time features from Session_Start	0-23, Mon-Sun, Jan-Dec

4. Method and tools

Python notebook (EV.ipynb)

- Loaded the raw file and inspected shape, data types, summary statistics, unique counts and missing values.
- Removed duplicate rows.
- Filled missing City and Charger_Type with the mode; filled missing Wait_Time_Min, Revenue_INR and Customer_Rating with the median.
- Reviewed outliers in the numeric columns and exported the result as EV_Cleaned.csv.

Power BI dashboard

Two pages were built on the cleaned table. **Page 1** is the Business Overview (sessions, revenue, energy, wait time, rating, sessions by hour, sessions by city, energy by vehicle type, revenue by charger type). **Page 2** is Operations and Customer Experience (charging duration, energy, revenue per session, rating, wait time by city, duration by charger type, revenue by city). Slicers for City, Station_Type, Vehicle_Type and Charger_Type work on both pages.

Extra checks done for this report

- Compared each category against its peers to look for gaps that are too large to be natural.
- Counted how many rows carry exactly the median value used for filling.
- Looked at the distribution of wait time, rating and revenue per kWh.
- Tested correlations between energy, duration, wait time, revenue and rating.

> **Why the extra checks matter:** a dashboard can show a clean-looking chart that is really the result of how gaps were filled. Testing for this before drawing conclusions is what separates reporting from analysis.

5. Data quality and cleaning

Cleaning was done in the notebook with a simple, common approach: remove duplicates, fill categorical gaps with the mode, and fill numeric gaps with the median. The median is a sensible choice for skewed numbers like wait time and revenue, because it is not pulled around by extreme values.

Step	What was done	Why
Duplicates	Dropped duplicate rows	Avoid counting a session twice
Missing City, Charger_Type	Filled with the most frequent value	Keeps all rows; categories cannot use a median
Missing Wait, Revenue, Rating	Filled with the median	Robust to outliers
Outlier review	Summary statistics and box plots	Spot impossible or extreme values
Time features	Created Hour, Day, Month	Needed for the time charts
Export	Saved EV_Cleaned.csv	Source for Power BI

The trade-off

Filling gaps keeps the row count high, but every filled cell is an invented value. When many gaps are filled with the same number, that number becomes over-represented. The next section shows that this is exactly what happened here: about 2,000 rows were affected in each of five columns.

What a better practice looks like

- Add a flag column (for example City_Imputed = 1) so filled rows can be excluded or tracked.
- For categories, use "Unknown" instead of the mode, so no real city or charger gets credit it did not earn.
- For revenue and ratings, consider leaving the value empty and reporting on valid rows only.
- Keep a short data-quality log with the count of rows changed in each column.

6. What the cleaning left behind

Five signs of filled or planted values

Several results in the dashboard look like business findings but trace back to the cleaning. Each of the five patterns below involves about 2,000 rows, which is exactly 2% of the data. That repeated number is the giveaway.

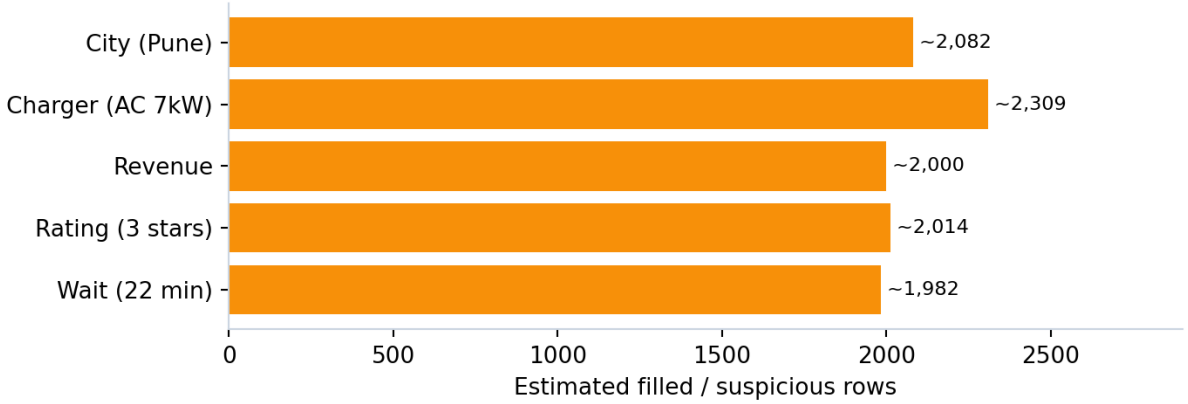


Figure 1. Estimated rows affected in each column (about 2,000 each).

Pattern	Evidence	Likely cause
Pune has the most sessions	14,322 vs about 12,240 for the other seven: about 2,080 extra	Missing City filled with the mode (Pune)
AC 7kW has the most sessions	26,732 vs about 24,420 for the other three: about 2,300 extra	Missing Charger_Type filled with the mode
Revenue stuck at one value	2,000 rows equal exactly Rs 781.435	Missing revenue filled with the median
Too many 3-star ratings	21,611 vs about 19,600 for each other star	Missing rating filled with the median (3)
Spike at 22 min wait	About 2,000 more than neighbouring minutes	Missing wait filled with the median (22)

Pune and AC 7kW: the "leaders" that probably are not

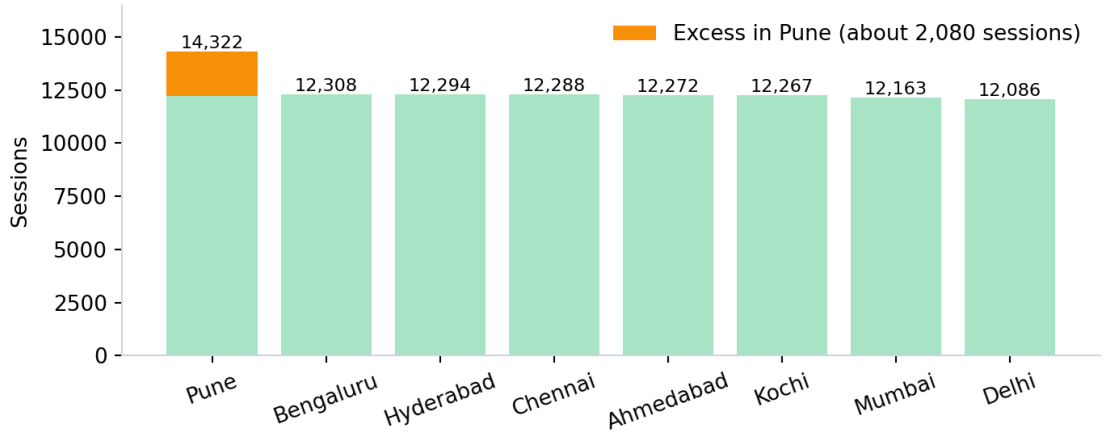


Figure 2. Sessions by city. The amber block is the surplus that matches the number of filled City values.

Pune also earns Rs 11.9M against about Rs 10.1M elsewhere. Roughly Rs 1.7M of that gap equals 2,000 sessions at the average Rs 823, which again points to filled rows and not stronger demand. Pune's revenue per session (Rs 828) and wait time (23.3 min) are ordinary.

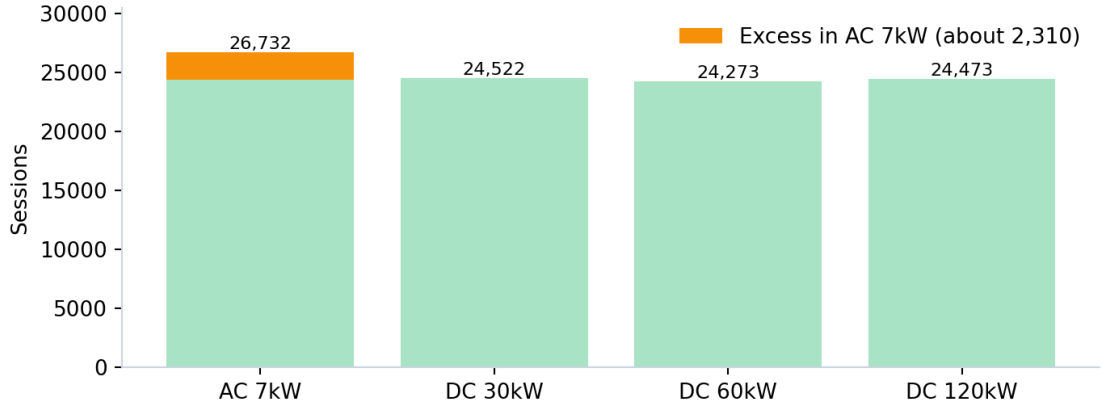


Figure 3. Sessions by charger type. The same effect appears for AC 7kW.

Other quality notes

- **500 extreme waits.** Exactly 500 sessions wait over 60 minutes (up to 219), while nobody waits between 46 and 60. These look like inserted outliers, not natural queueing.
- **2,146 sessions with a 0-minute wait,** about 2.1% of the data, another unusually round number.
- **Station_ID is not unique.** All 300 codes appear in every city and every station type, so a code cannot identify a physical station. Station-level analysis is therefore unreliable.
- **Charging duration ignores power.** A 120 kW charger takes the same time (about 65 min) as a 7 kW one, and duration is uncorrelated with energy (r = 0.002).

> **Take-away:** the structure of the data (uniform spreads, round counts, no cross-variable relationships) strongly suggests a synthetic or partly simulated dataset. The analysis is still a valid exercise, but conclusions should be described as patterns in this data, not as proven facts about a real network.

7. Headline KPIs

Matches the Power BI dashboard values

100K Total sessions	**Rs 82.30M** Total revenue	**5.15M kWh** Total energy	**23.31 min** Average wait
3.00 Average rating	**64.90 min** Avg charging duration	**51.47 kWh** Avg energy / session	**Rs 822.96** Avg revenue / session

Metric	Mean	Median	Min	Max
Energy (kWh)	51.47	51.43	8.01	95.00
Wait time (min)	23.31	22	0	219
Charging duration (min)	64.90	65	10	120
Revenue (Rs)	822.96	781.44	80.79	2,082.29
Customer rating	3.00	3	1	5

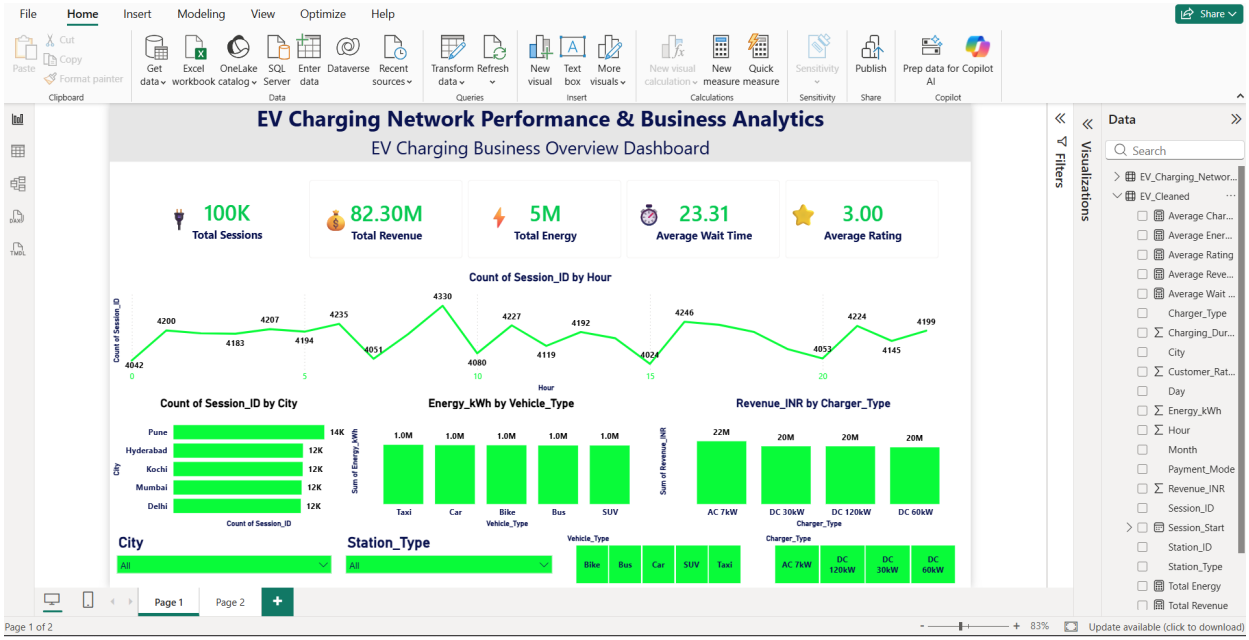
Reading the KPIs

- **Price level:** total revenue divided by total energy is about Rs 16 per kWh, and individual sessions range between about Rs 8 and Rs 19 per kWh.
- **Duration is capped:** sessions last 10 to 120 minutes, averaging just under an hour.
- **Mean vs median wait:** the mean wait (23.3) is slightly above the median (22) because of the 500 extreme waits.
- **Rating of 3.00 is exactly the middle of the scale,** which fits evenly spread ratings rather than a measured opinion.

> The KPIs are the same on both dashboard pages, so the report and the dashboards agree with each other.

8. Power BI dashboard walkthrough

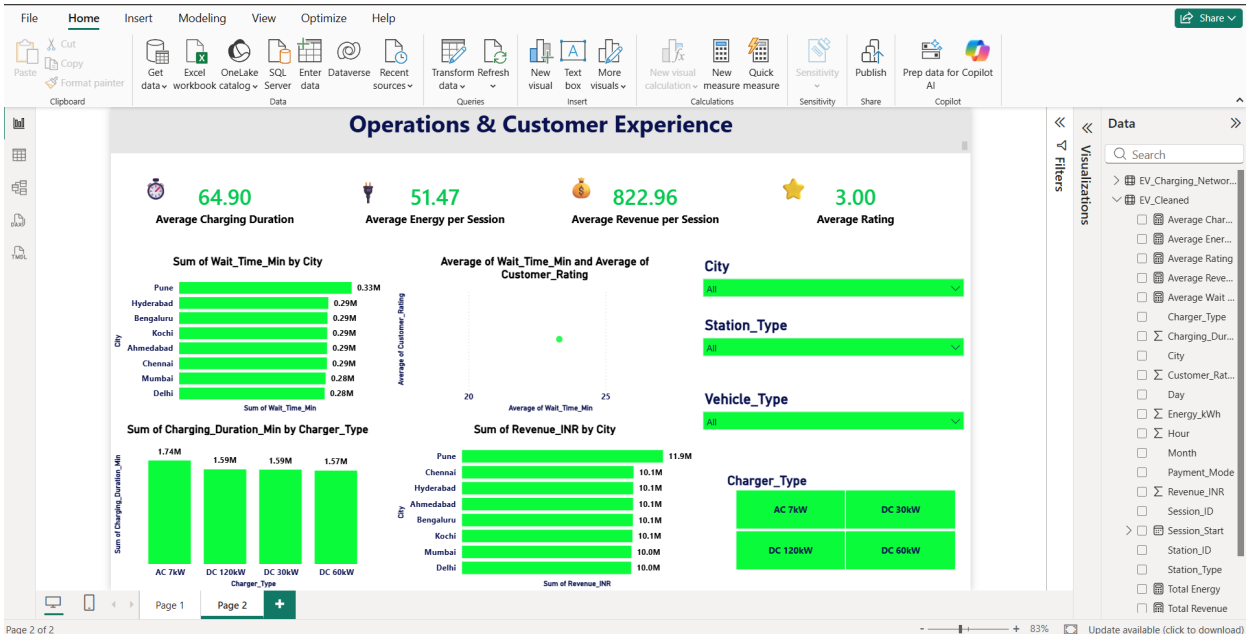
Page 1: Business Overview



The first page gives a quick health check: five KPI cards, an hourly line chart, sessions by city, energy by vehicle type and revenue by charger type, with slicers for City, Station_Type, Vehicle_Type and Charger_Type.

- **Strength:** a clear layout with one idea per chart and working slicers.
- **Improve:** the bar charts start at zero, so tiny differences look like none at all; add data labels for percentage difference or a "vs average" measure.
- **Improve:** the green on green slicers are hard to read; use a neutral slicer colour and keep green for data.

Page 2: Operations and Customer Experience



The second page focuses on operations: charging duration, energy and revenue per session, rating, wait time by city, duration by charger type and revenue by city.

- **Scatter chart shows one dot.** The wait-versus-rating chart plots only the overall average, so it says nothing about the relationship. Put Station_ID or Session_ID in the Details field to see the real spread.
- **"Sum of Wait_Time_Min" is the wrong measure.** Summing minutes (0.33M for Pune) mostly reflects session count. Use the average wait instead.
- **Same issue for duration:** use average charging duration by charger type, which is almost identical (about 65 min) for all four.

9. When people charge

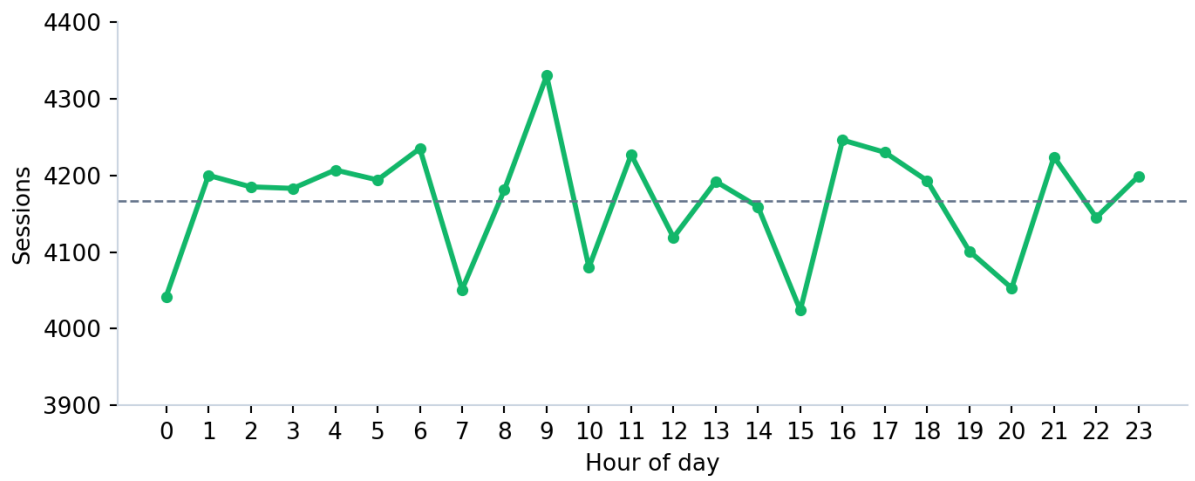


Figure 4. Sessions by hour of day; the dashed line is the hourly average (about 4,170).

Every hour of the day receives between 4,024 and 4,330 sessions. The peak at 09:00 is only about 8% above the quietest hour (15:00). There is no morning or evening rush, no overnight dip, and revenue by hour follows the same flat pattern (Rs 3.3M to 3.5M per hour).

What this means

- **Capacity planning:** demand is spread over 24 hours, so chargers are used steadily and queues do not come from a rush.
- **Pricing:** time-of-day pricing would shift little behaviour here, since there is no peak to flatten.
- **Real-world check:** actual networks usually show clear commute-time and evening peaks. A flat curve is another sign the data is simulated, so check this against live station logs before applying it to a real network.
- **Wait is flat too:** average wait by hour stays between 22.5 and 23.7 minutes. Only 09:00 is slightly better (22.5), even though it is the busiest hour.

> **Insight:** wait time does not rise when more people charge. That means waits are not caused by demand peaks in this data, which weakens the usual fix of "add capacity at peak time".

10. Seasonality and year-on-year

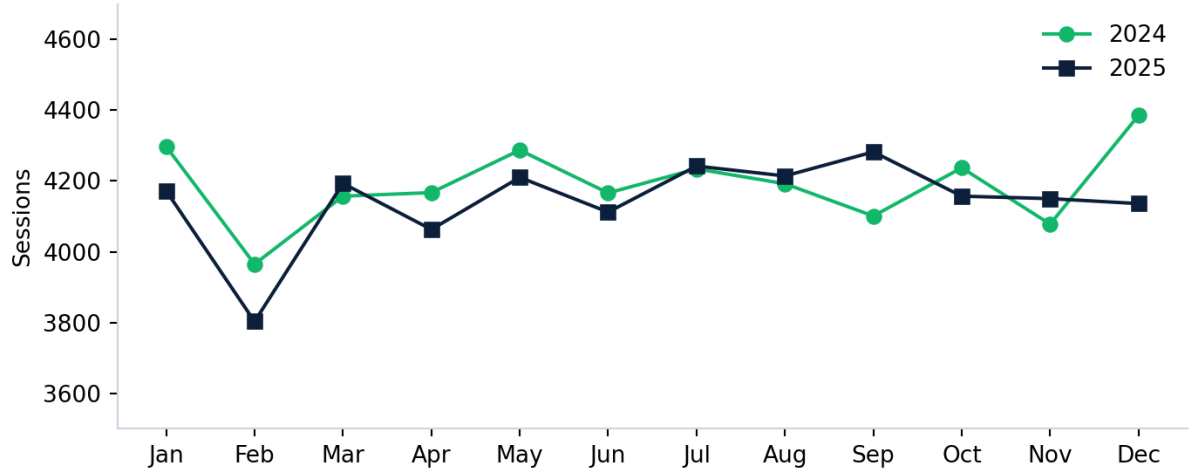


Figure 5. Sessions per month, 2024 vs 2025.

Year	Sessions	Revenue	Energy
2024	50,266	Rs 41.37M	2.59M kWh
2025	49,734	Rs 40.93M	2.56M kWh
Change	-1.1%	-1.1%	-1.1%

The network did not grow between the two years; volumes slipped about 1%. February is the lowest month (7,768 sessions) mostly because it has fewer days, and the highest months (December, May, July, January) are within 10% of the lowest.

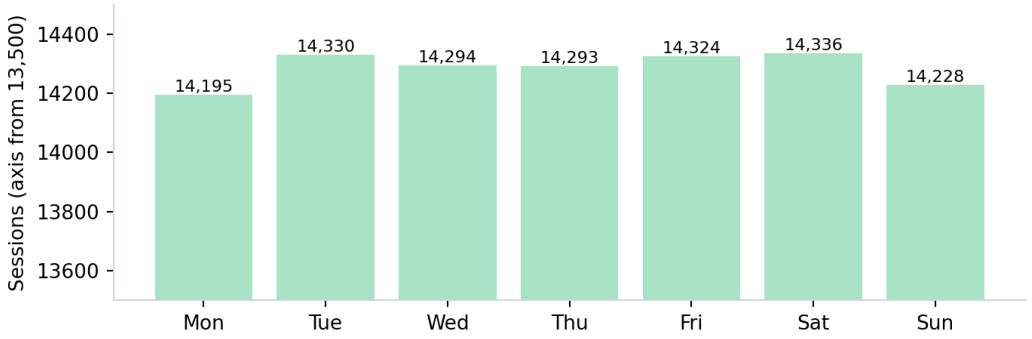


Figure 6. Sessions by weekday.

Weekdays differ by less than 1%: Saturday is highest (14,336) and Monday lowest (14,195). There is no weekend effect.

> **Insight:** a flat year-on-year result means growth has to come from new capacity, new customers or better pricing; there is no seasonal tailwind to rely on.

11. City analysis

City	Sessions	Revenue (Rs M)	Avg wait (min)	Avg rating	Revenue / session
Pune	14,322	11.86	23.29	3.02	828
Bengaluru	12,308	10.08	23.31	2.99	819
Hyderabad	12,294	10.10	23.47	3.00	822
Chennai	12,288	10.12	23.25	3.00	824
Ahmedabad	12,272	10.10	23.30	2.97	823
Kochi	12,267	10.06	23.34	3.02	820
Mumbai	12,163	10.00	23.29	2.99	822
Delhi	12,086	9.98	23.27	3.00	826

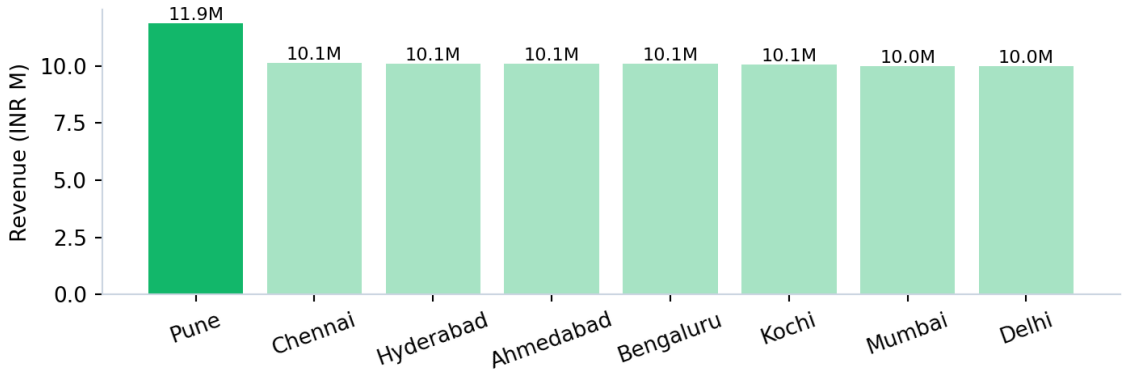


Figure 7. Revenue by city (Pune includes the filled rows).

- **The seven non-Pune cities are within 1.5% of each other** in sessions and revenue.
- **Per-session figures are identical everywhere:** wait 23.3 min, rating 3.0, revenue Rs 819 to 828. This is the clearest evidence that city does not change how the business performs.
- **Hyderabad,** the target market for this analyst, sits mid-table: 12,294 sessions, Rs 10.1M revenue, 23.5 min wait (the highest of the eight, by a hair).
- **Delhi** has the fewest sessions (12,086) but the highest revenue per session outside Pune.

Insight: there is no evidence for favouring one city. Hold off on the idea of copying Pune until a clean City column confirms that its lead is real.

12. Charger types

Charger	Sessions	Revenue (Rs M)	Avg duration (min)	Avg energy (kWh)	Revenue / session
AC 7kW	26,732	21.99	65.1	51.4	823
DC 30kW	24,522	20.22	64.7	51.6	825
DC 120kW	24,473	20.22	65.0	51.8	826
DC 60kW	24,273	19.86	64.7	51.1	818

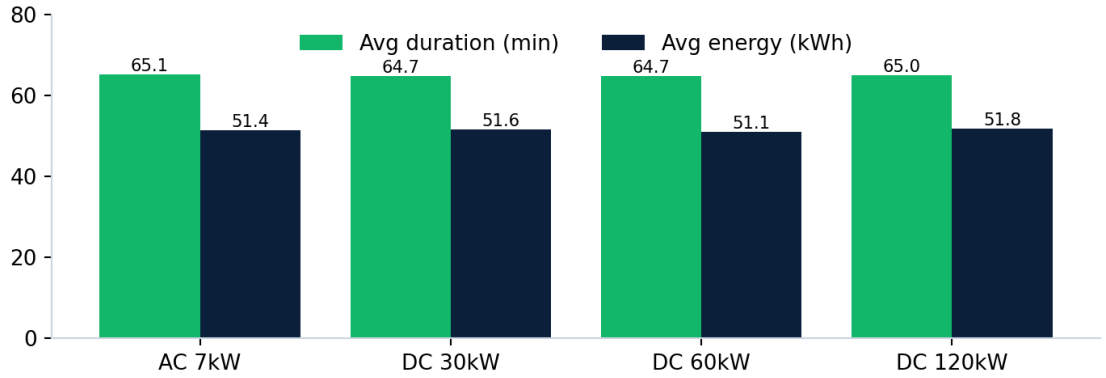


Figure 8. Average duration and energy by charger type.

- **All four chargers deliver the same energy (about 51 kWh) in the same time (about 65 minutes).** Physically, a 120 kW charger should deliver 51 kWh in about 25-30 minutes, and a 7 kW one in over 7 hours.
- **Revenue per session is the same (Rs 818-826),** so there is no price premium for fast charging in this data.
- **AC 7kW leads on sessions** only because of the roughly 2,300 filled rows (Figure 3). Excluding them, all four types carry about 24,400 sessions each.

Insight: if real, fast DC chargers are not earning more per session or saving customers time. That would argue for a tariff that rewards speed. If not real, the duration field needs to be fixed at source.

13. Vehicle types

Vehicle	Sessions	Revenue (Rs M)	Energy (M kWh)	Avg kWh / session	Avg rating
Taxi	20,157	16.61	1.037	51.5	2.97
Bike	20,052	16.44	1.029	51.3	3.01
Car	19,954	16.43	1.031	51.7	3.01
Bus	19,933	16.42	1.025	51.4	3.01
SUV	19,904	16.39	1.025	51.5	3.00

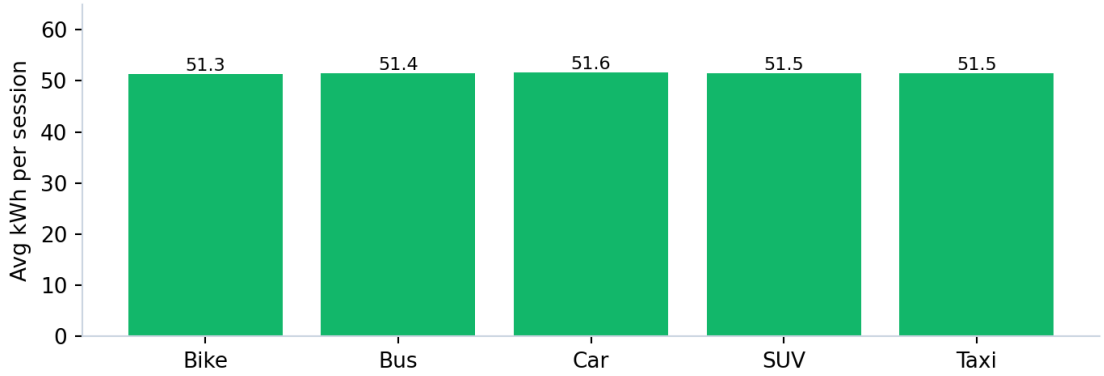


Figure 9. Average energy per session by vehicle type.

Each vehicle category supplies about one fifth of sessions and one fifth of revenue. Taxis are marginally ahead in sessions (20,157) but also have the lowest average rating (2.97), though the difference is too small to act on.

A reality check

A two-wheeler battery holds a few kWh, while a bus can take 200+ kWh. Here a bike takes 51.3 kWh on average and a bus 51.4. Energy per session does not depend on vehicle type at all. For a real operator this would be a serious data flaw, because vehicle mix is a key planning input.

> **Insight:** the vehicle segments are equally important to the business as recorded. Recommended next step is to confirm whether Vehicle_Type is recorded at the charger or entered manually.

14. Station types and stations

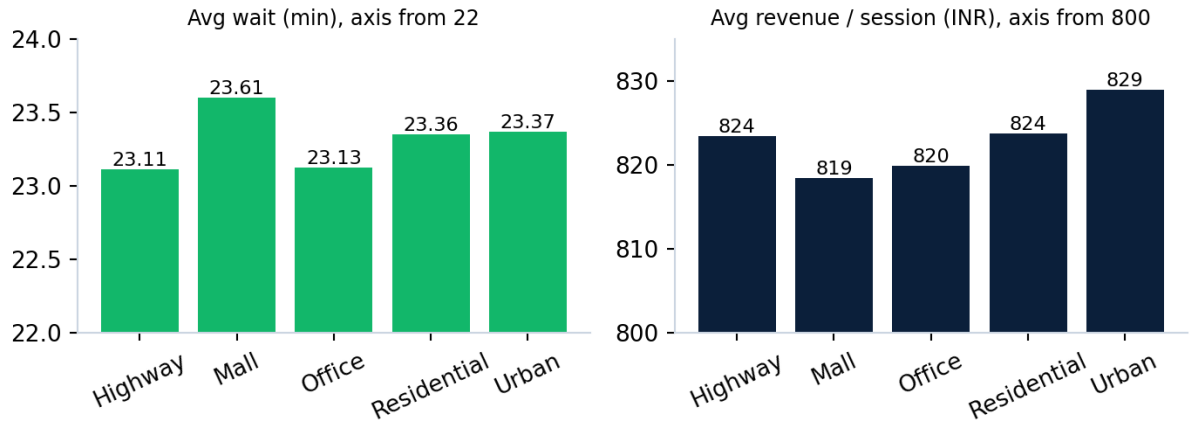


Figure 10. Station-type comparison (note the truncated axes).

Station type	Sessions	Avg wait (min)	Revenue / session	Avg rating
Highway	20,261	23.11	823.5	2.99
Office	20,010	23.13	820.0	2.99
Residential	19,937	23.36	823.8	3.01
Urban	19,901	23.37	829.0	3.01
Mall	19,891	23.61	818.5	3.00

Mall stations have the longest wait (23.6 min) and the lowest revenue per session (Rs 818.5); Urban has the highest revenue per session (Rs 829). The spread is half a minute and Rs 10, so these are tiny.

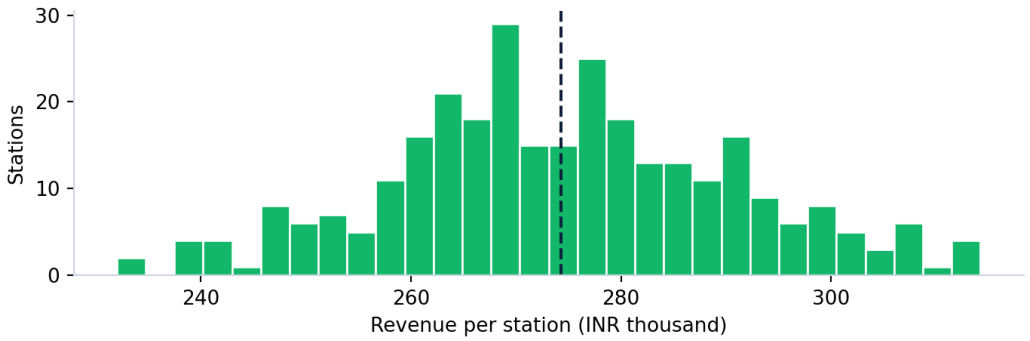


Figure 11. Revenue per Station_ID (300 IDs).

Station revenue ranges from about Rs 232K to Rs 314K and is bell-shaped around Rs 274K. Because each Station_ID appears in all eight cities and all five station types, this chart mixes unrelated sites and should not be used to rank real stations.

15. Payment behaviour

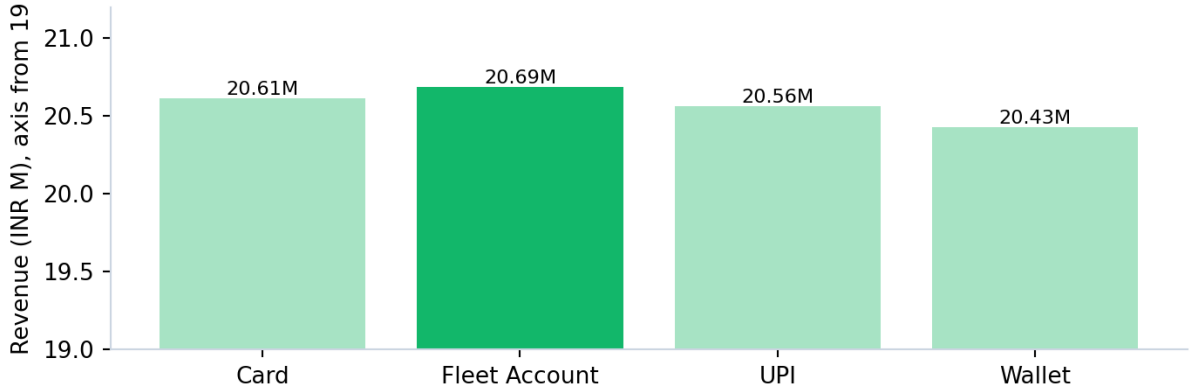


Figure 12. Revenue by payment mode.

Payment mode	Sessions	Share	Revenue (Rs M)	Revenue / session	Avg duration (min)
Card	25,122	25.1%	20.61	820.5	64.7
Fleet Account	24,999	25.0%	20.69	827.6	64.9
UPI	24,959	25.0%	20.56	823.9	64.8
Wallet	24,920	24.9%	20.43	819.9	65.2

- **All four modes hold a quarter of sessions each.** The mix of card, UPI, wallet and fleet is unusually even.
- **Fleet Account has the highest revenue per session (Rs 828)** and the highest total revenue (Rs 20.69M), but only by Rs 7 over Wallet.
- **Fleet customers are a stable B2B base:** 25% of volume likely comes from a small number of corporate accounts, which gives bargaining power for contracts and volume discounts.

> **Insight:** payment mode is not a lever for revenue. It is more useful operationally: wallet and fleet accounts allow pre-paid, contract and app-based flows that make billing and loyalty easier.

16. Revenue drivers and pricing

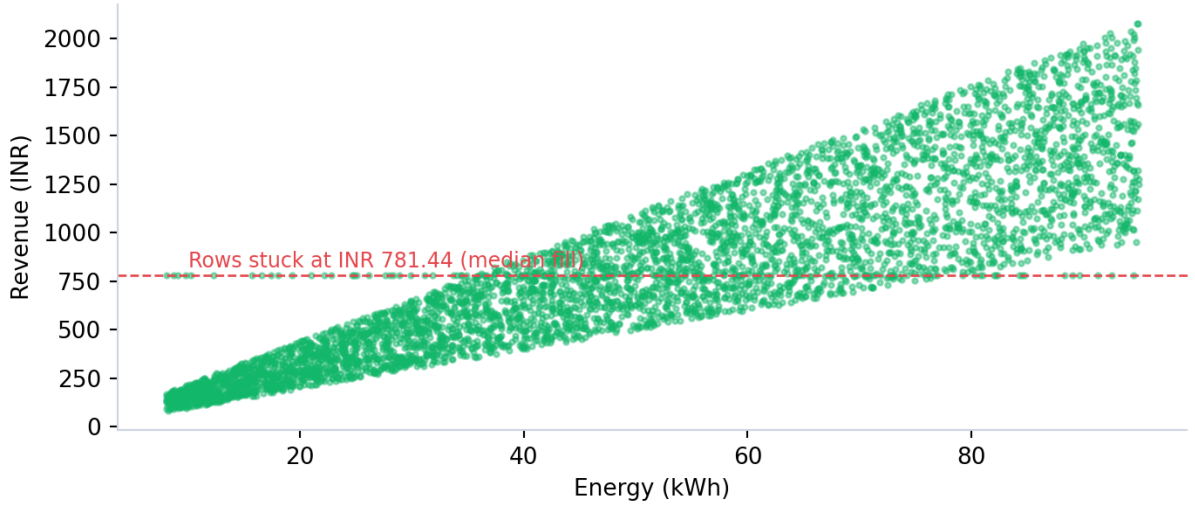


Figure 13. Energy vs revenue for a 5,000-session sample.

Revenue increases with energy (correlation 0.89), and each session falls inside a wedge between about Rs 8 and Rs 19 per kWh. The flat red line is the 2,000 rows whose revenue was replaced with the median value.

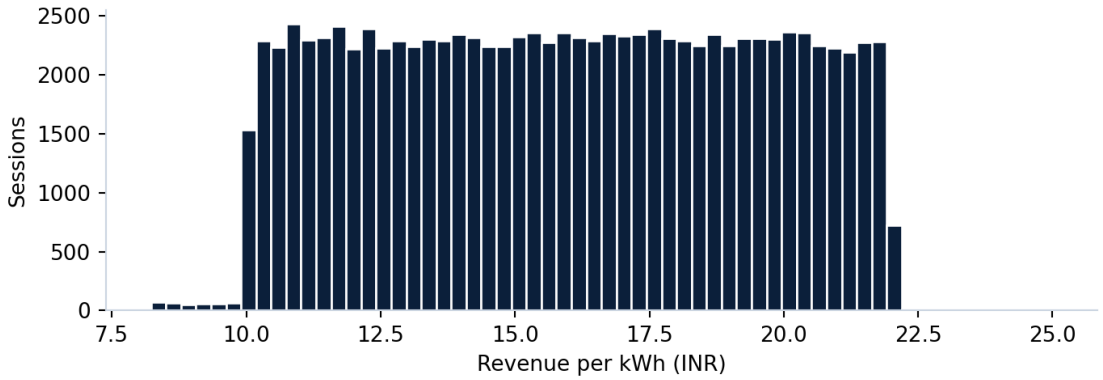


Figure 14. Distribution of revenue per kWh (values above Rs 25 hidden).

Findings

- **Price per kWh is roughly uniform between Rs 8 and Rs 19,** averaging Rs 16.1, and does not differ by charger type (Rs 16.10-16.16).
- **No relationship with city, vehicle, station or payment mode.** The only real driver is how much energy is delivered.
- **Pricing is not tiered.** A fast DC charger earns the same per kWh as a slow AC one, so there is room for premium pricing for speed and convenience if the market accepts it.
- **Revenue is not related to wait time or rating** (both r = 0.00). Slow service does not appear to cost money, and good service does not earn more.

> **Insight:** to grow revenue, raise the energy delivered per session or the price per kWh. Moving sessions between cities or hours will not change revenue.

17. Wait time

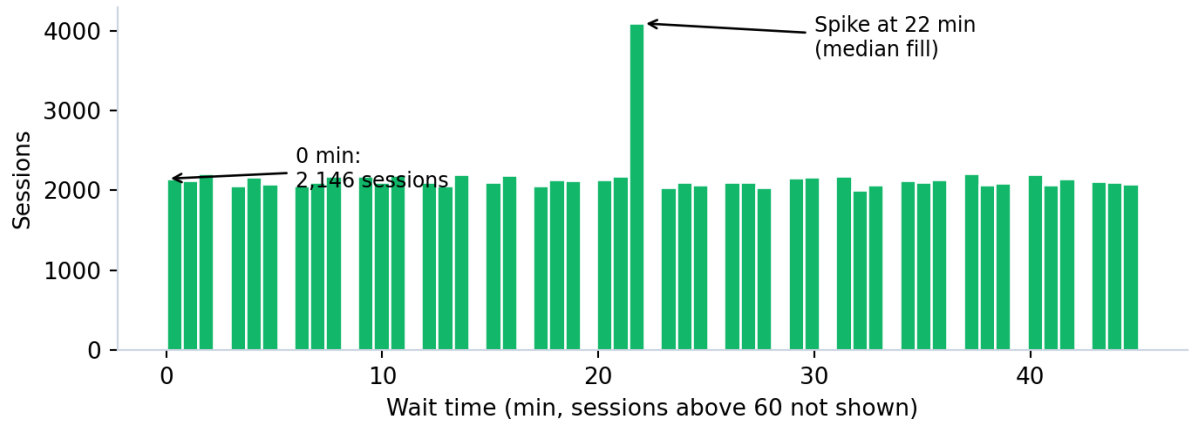


Figure 15. Distribution of wait time (0 to 60 minutes).

Statistic	Value
Mean / median	23.3 / 22 min
75th / 90th / 95th / 99th percentile	34 / 41 / 43 / 45 min
Sessions waiting over 30 minutes	about 32%
Sessions waiting over 45 minutes	500 (0.5%)
Maximum	219 min

- **Most people wait up to 45 minutes.** A long wait before a one-hour charge is a poor experience: customers spend nearly 40% as long queueing as charging (23 / 65 min).
- **Wait is the same everywhere:** by city (23.25 to 23.47), station type (23.1 to 23.6), vehicle (23.2 to 23.4) and hour (22.5 to 23.7). No location is the problem; the whole network is.
- **Long-tail waits (above 60 min) affect exactly 500 sessions** and appear as inserted outliers. Treat them separately in reports.
- **2,146 zero-minute waits** may be real walk-ups or missing data; check how the system records them.

Insight: with demand flat across 24 hours and wait equal in all segments, the cause is probably structural: too few connectors per station for the session length, and every session being about an hour long.

18. Customer ratings

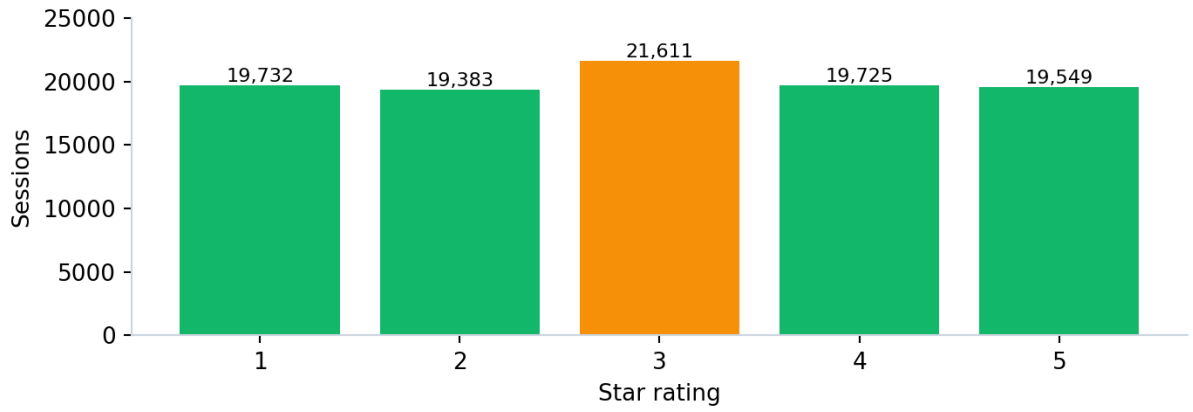


Figure 16. Ratings; the amber bar is inflated by median-filled rows.

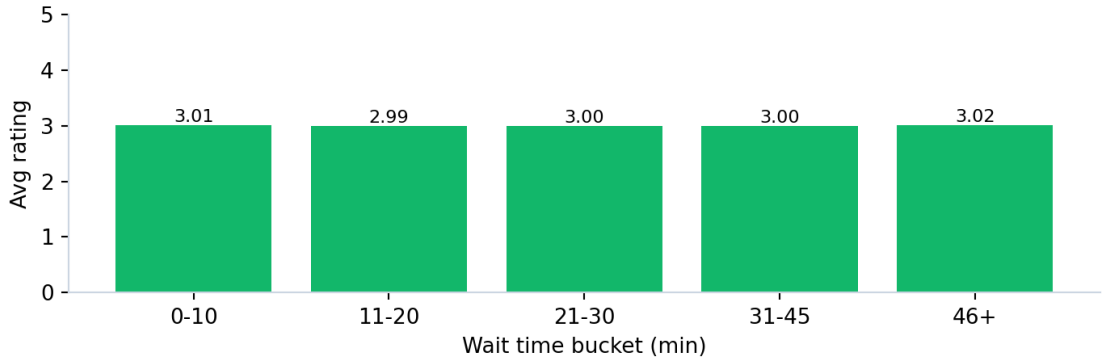


Figure 17. Average rating by wait-time bucket.

Ratings are spread evenly: about 39% of sessions give 1 or 2 stars and about 39% give 4 or 5. The average rating by wait bucket stays at 3.0 whether customers wait 5 minutes or 45. Ratings are also unrelated to city, charger, vehicle, station type, payment mode and year (2024: 3.00, 2025: 3.00).

What this implies

- **The rating tells us nothing about service quality in its current form.** Normally long waits depress ratings; here they do not.
- **One in five customers is unhappy (1-2 stars) in every segment,** so dissatisfaction is a network-wide issue, not a local one.
- **Without reason codes the rating cannot be acted on.** Customers give a score but not the cause (wait, price, charger fault, app).

> **Insight:** before using ratings for decisions, check how they are collected (when the prompt appears, whether it is optional, and whether it is linked to the right session).

19. Consolidated insights

#	Insight	Evidence	Confidence
1	Performance is balanced across all segments	All categories within 1-2%; per-session metrics identical	High
2	Pune and AC 7kW "leads" are caused by filling gaps	About 2,000 extra rows each, matching the fill	High
3	Revenue depends on energy delivered only	r = 0.89; Rs 16 per kWh in every segment	High
4	Wait time averages 23 minutes everywhere	Flat by city, hour, station, vehicle	High
5	Ratings do not respond to wait or any factor	r = 0.00; flat across buckets	High (data issue likely)
6	No demand peak across the day	4,024-4,330 sessions in every hour	Medium (looks simulated)
7	Volumes fell about 1% from 2024 to 2025	50,266 vs 49,734 sessions	Medium
8	Fast chargers earn no premium and take no less time	Same duration, price per kWh and revenue	Medium (check source)
9	Fleet accounts are the top revenue payment mode	Rs 20.69M; Rs 828 per session	Low (gap is small)
10	Station_ID cannot identify unique stations	Every ID appears in all 8 cities	High

Insights 1 to 5 are the safest to share with a business audience. Insights 6 to 8 depend on whether the data represents real behaviour, and should be confirmed with source systems before decisions are made.

20. Recommendations

Prioritised by impact and effort

Priority	Recommendation	Why	Success measure
1 (high)	**Cut average wait from 23 to under 15 minutes.** Add connectors or a booking/reservation option; add live queue status in the app.	Wait is the biggest, network-wide experience gap and is almost 40% of charge time	Avg wait, share of sessions above 30 min
2 (high)	**Fix the data pipeline.** Stop filling City and Charger_Type with the mode; use "Unknown"; add an imputation flag column; log changes.	About 2,000 rows each in five columns are distorting rankings	Imputed rows below 0.5%
3 (high)	**Make ratings meaningful.** Link each rating to a session, ask for a reason (wait, price, charger fault, app) and make the prompt consistent.	Ratings show no link to any factor	Rating response rate; correlation with wait
4 (medium)	**Introduce speed-based pricing.** A premium for DC 60/120 kW, and a lower price for slow AC.	Price per kWh is identical across chargers	Revenue per kWh by charger
5 (medium)	**Raise revenue per session through energy,** with bundles or subscriptions that encourage fuller charges.	Revenue follows kWh only	kWh per session, revenue per session
6 (medium)	**Develop fleet contracts** with taxi and bus operators: volume tariffs, priority lanes, monthly invoicing.	Fleet is the top revenue mode and taxis are the top vehicle	Fleet share of revenue
7 (medium)	**Validate duration, energy and vehicle fields against charger logs.**	120 kW and 7 kW show the same duration; bike and bus the same energy	Duration by charger matches spec
8 (low)	**Hold off on location-based investment** until the City and Station_ID columns are clean.	Rankings are driven by filled values	Unique station IDs per city

Quick wins for the next 30 days

- Re-run the Power BI averages instead of sums on Page 2 (wait time, duration).
- Fix the scatter chart so it plots every session or station rather than one dot.
- Add a "data quality" card to the dashboard showing the share of filled rows.
- Add a note on the dashboard that Pune and AC 7kW figures include filled values until the raw data is rechecked.

Dashboard improvements

- Add a "vs average" measure and conditional colour, so small gaps are shown honestly instead of looking like equal bars.
- Use neutral slicer colours; keep green for the main data series.
- Add a third page: Data Quality and Pricing (revenue per kWh, imputation counts, station-ID checks).
- Add year and month slicers to compare 2024 and 2025.

Expected impact

Action	Effect on business
Reduce wait by 8 minutes	Customers spend about 12% less total time at the station; likely better ratings once they are measurable
Speed-based pricing	Even a Rs 1 per kWh premium on DC sessions (about 49K sessions, 51 kWh each) would add roughly Rs 2.5M a year
Fleet contracts	Locks in a quarter of volume and raises predictability
Clean data	Removes the risk of investing based on a false city or charger ranking

The pricing estimate is illustrative: it multiplies the observed DC session count by average energy per session and ignores any change in demand.

21. Limitations, next steps and conclusion

Limitations

- The dataset appears synthetic or partly simulated, so patterns may not match a real network.
- About 2% of rows in five columns were filled in; their effect was estimated but not removed.
- No costs, tariffs, station capacity or customer IDs, so profit, utilisation and repeat behaviour cannot be measured.
- Station_ID is not unique, so station-level analysis is unreliable.
- The raw file was not available for this report, so the filled-row counts are inferred from the cleaned data and the notebook logic.

Next steps

- Re-check the raw data to confirm which rows were filled and remove them from the analysis.
- Rebuild the dashboard using clean City and Charger_Type values and average measures.
- Add capacity and cost data to move from revenue analysis to profitability.
- Run a customer survey to understand the drivers of dissatisfaction.

Conclusion

> The EV charging network in this dataset is large, stable and remarkably uniform: Rs 82.3M from 100,000 sessions, with almost no difference between cities, chargers, vehicles, stations or payment methods. The results that look like standouts (Pune and AC 7kW) come from how missing values were filled. The real opportunities lie in cutting a 23-minute average wait, collecting ratings that mean something, pricing for speed, and cleaning the data so that future decisions rest on solid numbers.

Prepared by Syed. Data: EV_Cleaned.csv (100,000 sessions). Tools: Python and Power BI.